



AI ASSIGNMENT

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6TH SEMESTER – SEC 'C'
DEPARTMENT: BSCS

ASSIGNMENT:

AGENT: “EMPTY CLASSROOM FINDING AGENT”

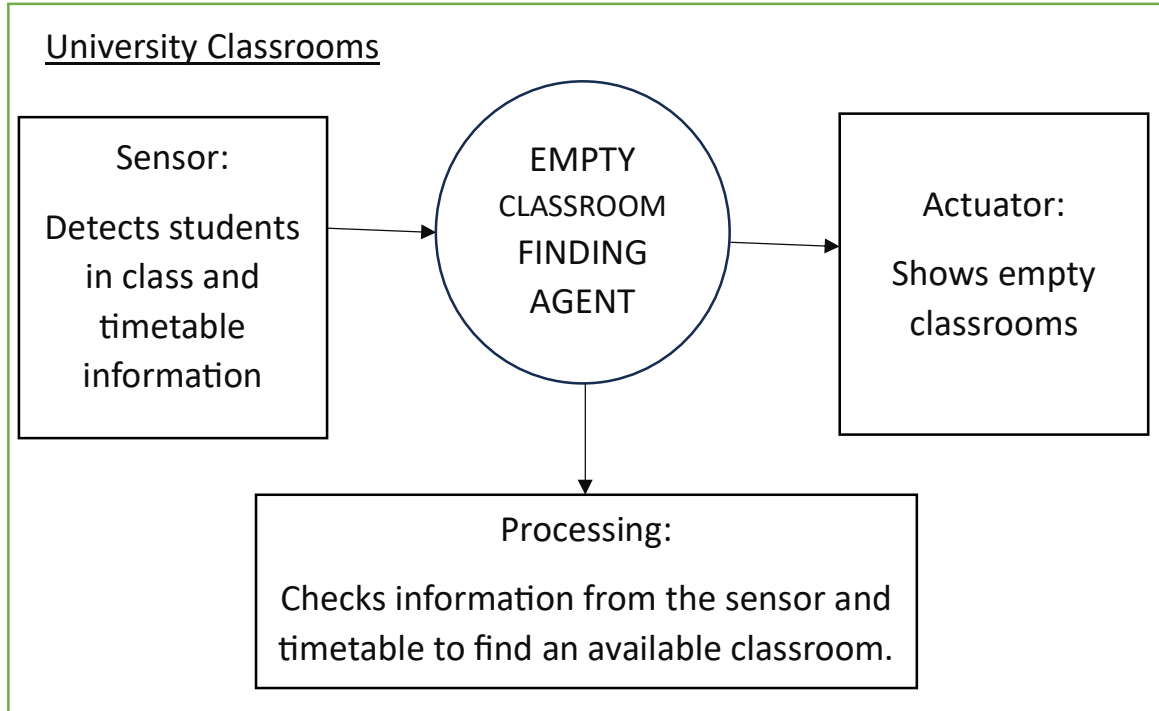
Q: Think of an AI agent and formulate it. Also evaluate your agent in PEAS Framework.

- **Agent Name:** “Empty Classroom Finding Agent”.
- **Problem Formulation:** Students face difficulty in finding empty classrooms in university for studying, group discussions, etc.
- **Environment:** University class room.
- **Environment Types:**

<u>Property</u>	<u>Type</u>	<u>Why?</u>
Observability	Partially Observable	The agent may not know everything happening inside every classroom.
Determinism	Stochastic	Classroom availability can be uncertain; for example, a scheduled class may be cancelled without updating the timetable.
Episodes	Sequential	Current classroom information can affect future decisions about which room to recommend.
Change	Dynamic	Classroom status changes when students or teachers enter or leave.
Values	Discrete	Classroom status can be represented as Empty / Occupied.
Agents	Multi-Agent	Students, teachers, and staff can affect the classroom environment.

- **Agent Work:** The agent finds and empty and available classroom for students.

- **Graphical Representation:**



- **Sensor:**
 - i. **Occupancy Sensor:** Detects whether people are present in a classroom.
 - ii. **Timetable Information:** Provides information about scheduled classes.
- **Processing:** The agent checks the occupancy data and timetable information to determine which classroom is available.
- **Actuator:** It displays the available classroom.

- **Peas Framework Evaluation:**

<u>PEAS</u>	<u>Description</u>
P – Performance	Accuracy: Correctly finds an empty classroom. Efficiency: Finds the classroom quickly.
E – Environment	University and classrooms.
A – Actuator	Suggest empty classrooms.
S – Sensor	Detects students in class and timetable information.